

STEM Course Prep
In Class Activities
Chapter 4

Slide 2

1. The following statements assign logical expressions to the variable x. Before putting the equations in MATLAB, write down the value of x. Check your answers in MATLAB.

```
X = 3>2
X = 2>3
X = -4 <= -3
X = 2~=2
X = 3 == 3
X = 0 < 0.5 < 1
```

Slides 3-5

2. Have MATLAB generate a random integer in the range of [-5, 5]. Use an IF block to determine if the number is greater than 1. If it is greater print out the following message: 'Number is greater than 1'.

```
Number=randi([-5,5])
if Number>1
    disp('Number is greater than 1');
end
```

3. Create a temperature conversion script that will convert to Kelvins. The script should ask the user for a temperature and its scale (Fahrenheit or Celsius).

```
%This code will convert a temperature to the Kelvin scale
format longg
temp=input('Enter a temperature in Fahrenheit or Celsius: ')
scale=input('Enter the temperature scale (F or C) you are using: ','s')
if scale == 'C'
    temp_kelvin=temp+273.15;
    fprintf('Temp in Kelvins = %.2f\n',temp_kelvin)
end
if scale == 'F'
    temp_rankine=temp+459.67;
    temp_kelvin=temp_rankine/1.8;
    fprintf('Temp in Kelvins = %.2f\n',temp_kelvin)
end
```

4. Check to see if a quadratic function has repeated roots. Recall that for $ax^2 + bx + c = 0$, repeated roots will occur if $b^2 - 4ac = 0$ and $a \neq 0$. Have the user input values for the coefficients: a, b, c . Display a message if roots are not repeated. (Use values $a=1, b=2, c=1$ for repeated roots)

```
a=input('Enter coefficient a: ')
b=input('Enter coefficient b: ')
c=input('Enter coefficient c: ')
repeated=0;
if(b^2-4*a*c==0) & (a~=0)
    x=-b/(2*a);
    fprintf('The repeated root is: %.4f',x)
    repeated = 1;
end

if repeated ==0
    disp('Roots are not repeated')
end
```

Slides 6-9

5. You want to solve a quadratic equation. You want to prevent divide by zero errors. You also want to display the roots if possible, as well as label them as complex or real. Hardcode the values of a, b, c as shown below. Vary the values to see what the code does.

```
a=0;
b=2;
c=1;
d=b^2-4*a*c;
if a ~=0
    if d<0
        disp('Complex roots')
        x1=(-b+sqrt(d))/(2*a)
        x2=(-b-sqrt(d))/(2*a)
    else
        disp('Real roots')
        x1=(-b+sqrt(d))/(2*a)
        x2=(-b-sqrt(d))/(2*a)
    end
else
    disp('a=0 will result in divide by zero error')
end
```

6. A bank has a savings account where the interest rate varies depending on the balance. See the following chart:

Balance	Interest Rate
<\$5000	9%
\$5000 <bal<\$10000	12%
>\$10000	15%

Use the `elseif` format to determine a customer's new balance after one year. (in the code below I used the `rand` function to determine a starting balance)

```
bal=15000*rand
if bal<5000
    rate=0.09
elseif bal<10000
    rate=0.12
else
    rate =0.15
end
newbal=bal+rate*bal;
fprintf('The new balance is: $%.2f\n',newbal)
```

Slide 10

7. Write a script that will allow a user to input a length and its unit. The script will take this information and convert the length to centimeters. Allowable units are m ,mm, in. All other units will be considered unknown. Display an error message if an unaccepted unit is used.

```
length=input('Enter the length you want to convert to cm: ');
unit=input('Enter the units (m, mm, in): ','s')
switch unit
    case 'm'
        y=length*100;
        fprintf('%.2f m converted to cm is %.2f', length,y)
    case 'mm'
        y=length/10;
        fprintf('%.2f mm converted to cm is %.2f', length,y)
    case 'in'
        y=2.54*length;
        fprintf('%.2f inches converted to cm is %.2f', length,y)
    otherwise
        disp('units are not known-try again')
end
```

8. Randomly generate an integer from 0 to 9. Determine if the number is even, odd or zero.

```
d=randi([0,9])
switch d
    case {2,4,6,8}
        disp('Even number')
    case {1,3,5,7,9}
        disp('Odd number')
    otherwise
        disp('zero')
end
```

9. This script will calculate velocity after a given time period. The user will input initial velocity and acceleration. The time interval will be selected from a menu. The menu will have three choices: 10, 20, 30 seconds.

```
vel=input('Enter initial velocity in m/s: ');
accel=input('Enter acceleration in m/s^2: ');
time=menu('Pick the time interval','10 sec','20 sec','30 sec')
switch time
    case 1
        t=10
    case 2
        t=20
    case 3
        t=30
    otherwise
        disp('You didn''t select a time')
end
if time~=0
    v=vel+accel*t;
    fprintf('The final velocity after %2d seconds is %.2f m/s.\n',t,v)
else
    disp('Please select a time from the menu')
end
```

10. Plot $\sin(x)$ for 0-360 degrees (use 5 degree increments). Allow the user to customize the plots by selecting color and plot markers from a menu.

```
x=0:5:360;
y=sind(x);
color=menu('Choose a plot color', 'Red','Magenta','Cyan','Yellow','Black')
marker=menu('Choose a plot marker','*','-'.')
switch color
    case 1
        if marker ==1
            plot(x,y,'r*')
        else
            plot(x,y,'r-'.')
        end

    case 2
        if marker ==1
            plot(x,y,'m*')
        else
            plot(x,y,'m-'.')
        end

    case 3
        if marker ==1
            plot(x,y,'c*')
        else
            plot(x,y,'c-'.')
        end
end
```

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```
case 4
    if marker ==1
        plot(x,y, 'y*')
    else
        plot(x,y, 'y-.' )
    end
case 5
    if marker ==1
        plot(x,y, 'k*')
    else
        plot(x,y, 'k-.' )
    end
otherwise
    plot(x,y)
end
```